* Derivation of the stresses and the tangent modulus for discontinuous damage

$$\psi = \frac{1}{2} K (\mathrm{tr}\,\underline{\varepsilon})^2 + (1-D)\psi^0 \qquad \psi^0 = \mu \|\mathrm{dev}\,\underline{\varepsilon}\|^2$$

$$\underline{\sigma} = \frac{\partial \psi}{\partial \underline{\varepsilon}} = K\,\mathrm{tr}\,\underline{\varepsilon}\,\underline{I} + (1-D)\,2\mu\,\mathrm{dev}\,\underline{\varepsilon}$$

$$\Delta\underline{\sigma} = \underline{\mathbb{C}} : \Delta\underline{\varepsilon}$$

$$\Delta\underline{\sigma} = \frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}} : \Delta\underline{\varepsilon} + \frac{\partial \underline{\sigma}}{\partial D}\frac{\partial D}{\partial \alpha} \otimes \frac{\partial \alpha}{\partial \underline{\varepsilon}} : \Delta\underline{\varepsilon}$$

$$\underline{\mathbb{C}} = \frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}} + \frac{\partial \underline{\sigma}}{\partial D}\frac{\partial D}{\partial \alpha} \otimes \frac{\partial \alpha}{\partial \underline{\varepsilon}}$$

$$\frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}} = K\,\underline{I}\otimes\underline{I} + (1-D)\,2\mu\,\underline{\mathbb{P}}$$

$$\frac{\partial \underline{\sigma}}{\partial D} = -2\mu\,\mathrm{dev}\,\underline{\varepsilon}$$

$$\frac{\partial D}{\partial \alpha} = \text{given}.$$

$$\psi^0_{n+1} = \mu \|\mathrm{dev}\,\underline{\varepsilon}_{n+1}\|^2$$

$$\frac{\partial \alpha_{n+1}}{\partial \underline{\varepsilon}_{n+1}} = 2\mu\,\mathrm{dev}\,\underline{\varepsilon}_{n+1}$$

$$\alpha_{n+1} = \begin{cases} \psi^0_{n+1}, & \psi^0_{n+1} \geq \psi^0_{max} \\ \alpha_n, & \psi^0_{n+1} < \psi^0_{max} \end{cases}$$

$$\frac{\partial \alpha_{n+1}}{\partial \underline{\varepsilon}_{n+1}} = \begin{cases} 2\mu\,\mathrm{dev}\,\underline{\varepsilon}_{n+1}, & \psi^0_{n+1} \geq \psi^0_{max} \\ \underline{0}, & \psi^0_{n+1} < \psi^0_{max} \end{cases}$$

$$\underline{\mathbb{C}}_{n+1} = K\,\underline{I}\otimes\underline{I} + 2\mu(1-D)\,\underline{\mathbb{P}} + \begin{cases} -2\mu\,\mathrm{dev}\,\underline{\varepsilon}_{n+1}\dfrac{\partial D(\alpha_{n+1})}{\partial \alpha_{n+1}} \otimes 2\mu\,\mathrm{dev}\,\underline{\varepsilon}_{n+1}, & \dot{\psi}^0_{n+1} \geq \alpha_n \\ 0, & \psi^0_{n+1} < \alpha_n \end{cases}$$

$$-4\mu^2 \frac{\partial D}{\partial \alpha}\,\mathrm{dev}\,\underline{\varepsilon}_{n+1} \otimes \mathrm{dev}\,\underline{\varepsilon}_{n+1}$$

$K = 1 \qquad \mu = 1 \qquad \alpha_n = 1$

$D_\infty = 0.98 \qquad D_0 = 1$

$$D = D_\infty\left(1 - e^{\frac{-\alpha}{D_0}}\right) = 0.98\left(1 - e^{\frac{\alpha}{\tau}}\right)$$

$$\underline{\underline{\varepsilon}}_n = \begin{bmatrix} 0 & 0.1 & 0 \\ 0.1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \underline{\underline{\varepsilon}}_{n+1} = \begin{bmatrix} 0 & \tau & 0 \\ \tau & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \tau \in \{0.7, 0.8\}$$

→ Calculate $\underline{\underline{\sigma}}_{n+1}$ & α_{n+1} for continious & discontinuous damage.

a) Continuous damage

$$\psi = \frac{1}{2} K (tr\,\underline{\underline{\varepsilon}})^2 + (1-D)\psi^0$$

$$\dot{\alpha} = |\dot{\psi}^0|$$

$$\alpha_{n+1} = \alpha_n + |\psi^0_{n+1} - \psi^0_n|$$

$$\psi^0 = \mu \|dev\,\underline{\underline{\varepsilon}}\|^2$$

$$\psi^0_{n+1} = \mu \|dev\,\underline{\underline{\varepsilon}}_{n+1}\|^2$$

$$\psi^0_n = \mu \|dev\,\underline{\underline{\varepsilon}}_n\|^2$$

ψ^0_{n+1}, ψ^0_n

→ $\alpha_{n+1} = \alpha_n + |\psi^0_{n+1} - \psi^0_n|$

$dev\,\underline{\underline{\varepsilon}}_{n+1} = \underline{\underline{\varepsilon}}_{n+1}$

$dev\,\underline{\underline{\varepsilon}}_n = \underline{\underline{\varepsilon}}_n - \frac{1}{3} tr\,\underline{\underline{\varepsilon}}_n \underline{\underline{I}}$

$dev\,\underline{\underline{\varepsilon}}_n = \underline{\underline{\varepsilon}}_n$

$$\alpha_{n+1} = 1 + \left| \sqrt{\tau^2 + \tau^2} - \sqrt{0.1^2 + 0.1^2} \right|$$

$$= 1 + \left| 2\tau^2 - 2 \times 0.1^2 \right|$$

$\tau = 0.7$

$$= 1 + 2(0.7)^2 - 2(0.1^2)$$

$$= 1 + 2\left(0.7^2 - 0.1^2\right)$$

$$= 1 + 2(0.49 - 0.01)$$

$$= 1 + 2(0.48)$$

$$\alpha_{n+1} = 1 + 0.96$$

$$\alpha_{n+1} = 1.96$$

$$\underline{\underline{\sigma}}_{n+1} = K\,tr\,\underline{\underline{\varepsilon}}_{n+1}\underline{\underline{I}} + 2\mu\left(1 - D(\alpha_{n+1})\right) dev\,\underline{\underline{\varepsilon}}_{n+1}$$

($tr\,\underline{\underline{\varepsilon}}_{n+1} = 0$)

$$= 2 \times 1\left(1 - D(1.96)\right) \begin{bmatrix} 0 & 0.7 & 0 \\ 0.7 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= 2 \times (1 - 0.842) \begin{bmatrix} 0 & 0.7 & 0 \\ 0.7 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\underline{\underline{\sigma}}_{n+1} = \begin{bmatrix} 0 & 0.2218 & 0 \\ 0.2218 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\tau = 0.8$

$$= 1 + 2(0.8)^2 - 2(0.1)^2$$

$$= 1 + 2(0.8^2 - 0.1^2)$$

$$= 1 + 2(0.64 - 0.01)$$

$$= 1 + 2(0.63) = 2.26$$

b) Discontinuous damage.

$$\alpha_{n+1} = \begin{cases} \psi^0_{n+1} & , \ \psi^0_{n+1} \geq \alpha_n \\ \alpha_n & , \ \psi^0_{n+1} < \alpha_n \end{cases}$$

$\tau = 0.7$ $\quad \psi^0_{n+1} = 2\tau^2 > \alpha_n$ (1)

$2 \cdot (0.7)^2 = 2 \times 0.49 = 0.98 < 1$

$\alpha_{n+1} = \alpha_n = 1$

$t = 0.8$

$\psi^0_{n+1} = 2 \cdot (0.8)^2 = 2 \times 0.64$

$\psi^0_{n+1} = 1.28 > (\alpha_n = 1)$

$\alpha_{n+1} = \psi^0_{n+1} = 1.28$

$\underline{\sigma}_{n+1} = K \, tr \, \underline{\varepsilon}_{n+1} \, \underline{I} + 2\mu (1 - D(\alpha_{n+1})) \, dev \underline{\varepsilon}_{n+1}$